THE GENERIC POINTS OF THE COMPLEMENT OF AN AFFINE OPEN IMMERSION HAVE LOCAL RINGS OF DIMENSION AT MOST ONE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

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Editorial note. Statement span 3449-3460 and proof span 3462-3482 come from the halfA pool card operations/pool-cards/stacks-divisors-halfA.json and were re-read in the pinned chapter file before this card was written.

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Editorial note. The proof contains the author's parenthetical hint to use the statement on points specializing to a point; it is preserved verbatim.

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Allowed standard prerequisites (editorial).

- The points of the spectrum of a local ring correspond to the specializations of the point, so the fibre of an open immersion over the spectrum of a local ring at a generic point of the complement is the punctured spectrum. Cited at proof lines 3462-3467. Identifies the restriction of the open subscheme over the local ring with the punctured spectrum.
- Affineness is preserved under base change, cited from the morphisms chapter. Cited at proof lines 3468-3470. Makes the restricted open affine, which allows the characterization of punctured spectra of dimension at most one to be applied.
- The punctured spectrum of a Noetherian local ring is affine exactly when the dimension is at most one, established by a previous card of this lane. Cited at proof lines 3470-3473. Gives the claimed dimension bound.
- A locally Noetherian scheme locally has finitely many irreducible components, so the generic point of a component of the closure of the open that contains the given point lies in the open. Used at proof lines 3474-3481.

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Produces a specialization from the open to the point, showing that the local ring cannot have dimension zero when the point lies in the closure.

Source reference. Original lemma 0BCR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14289–14289. Same-chapter reference used in the retained author text at source line(s) 3420, 3441, 3474. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01SD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2334–2334. Same-chapter reference used in the retained author text at source line(s) 3471, 8930. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Source reference. Original lemma 01J7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2004–2004. Same-chapter reference used in the retained author text at source line(s) 3469. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

Lemma 0.1. *Let X be a locally Noetherian scheme. Let $U \subset X$ be an open subscheme such that the inclusion morphism $U \to X$ is affine. For every generic point ξ of an irreducible component of $X \setminus U$ the local ring $\mathcal{O}_{X,\xi}$ has dimension ≤ 1. If U is dense or if ξ is in the closure of U, then* $\dim(\mathcal{O}_{X,\xi}) = 1$.

Proof. Since ξ is a generic point of $X \setminus U$, we see that

$$U_\xi = U \times_X \mathrm{Spec}(\mathcal{O}_{X,\xi}) \subset \mathrm{Spec}(\mathcal{O}_{X,\xi})$$

is the punctured spectrum of $\mathcal{O}_{X,\xi}$ (hint: use Schemes, Lemma 01J7). As $U \to X$ is affine, we see that $U_\xi \to \mathrm{Spec}(\mathcal{O}_{X,\xi})$ is affine (Morphisms, Lemma 01SD) and we conclude that U_ξ is affine. Hence $\dim(\mathcal{O}_{X,\xi}) \leq 1$ by Lemma 0BCR. If $\xi \in \overline{U}$, then there is a specialization $\eta \to \xi$ where $\eta \in U$ (just take η a generic point of an irreducible component of $\overline{U}$ which contains ξ; since $\overline{U}$ is locally Noetherian, hence locally has finitely many irreducible components, we see that $\eta \in U$). Then $\eta \in \mathrm{Spec}(\mathcal{O}_{X,\xi})$ and we see that the dimension cannot be 0. □